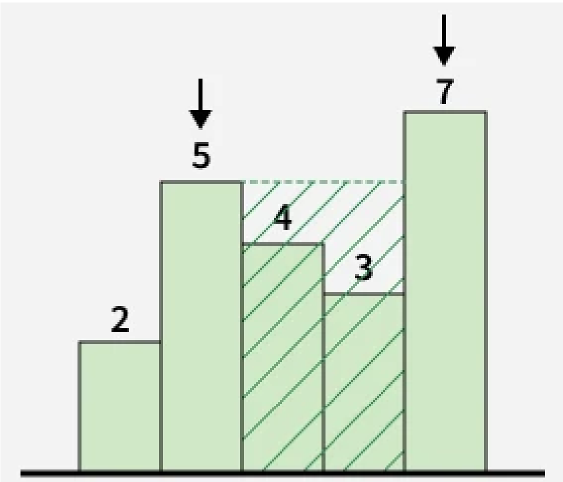
Maximum Area Between Bars

Difficulty: **Medium** Accuracy: **39.48%** Submissions: **21K+** Points: **4**

Given an integer array **height[]**, where height[i] represents the height of the i^{th} bar arranged in a row, find the **maximum** rectangular **area** that can be formed by selecting any two bars. The area is calculated based on the original positions of the selected bars.

Examples :

Input: height[] = [2, 5, 4, 3, 7]
Output: 10
Explanation:



The maximum rectangular area is formed by selecting the

Python3 Start Timer

```python
class Solution:
    def maxArea(self, height):
        n = len(height)
        left, right = 0, n - 1
        ans = 0

        while left < right:
            width = right - left - 1
            area = min(height[left], height[right]) * w
            ans = max(ans, area)

            if height[left] < height[right]:
                left += 1
            else:
                right -= 1

        return ans
```

bars of heights 5 and 7.
There are 2 bars between them, so the area is: min(5, 7)
2 = 10

Input: height[] = [1, 3, 4]
Output: 1
Explanation: Selecting bars 1 and 4 gives one bar between them, so the area is: min(1, 4) × 1 = 1

Constraints:
$1 \leq height.size() \leq 10^5$
$1 \leq height[i] \leq 10^4$

Try more examples

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